

INDEPENDENT UNIVERSITY BANGLADESH

Department of Computer Science and Engineering

SOFTWARE ENGINEERING PROJECT REPORT

Smart Healthcare Management System

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Abstract

Healthcare organizations need efficient methods for managing patient information, medical history, appointments, prescriptions, investigations, billing, insurance, symptoms, and preventive care. Manual record management can make information difficult to organize, search, update, and maintain. The Smart Healthcare Management System is a web-based application designed to provide a centralized platform for managing these healthcare activities.

The system provides modules for patient management, doctor management, appointment scheduling, medical records, prescription management, laboratory investigations, billing, insurance, symptom tracking, and preventive care. The application uses HTML5, CSS3, Bootstrap 5, JavaScript, PHP, MySQL, PDO, and Apache/XAMPP. A three-layer architecture separates the presentation, application, and database responsibilities.

The project demonstrates the practical application of software engineering concepts through requirement analysis, system design, database modeling, implementation, testing, security considerations, and maintenance planning. The final system provides a centralized approach to healthcare information management and reduces dependence on manually maintained records.

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Chapter 1

Introduction

Healthcare management is an important area in which accurate and organized information handling is necessary. Hospitals and clinics need to maintain patient information, medical history, appointment schedules, treatment records, prescriptions, investigations, billing information, and insurance records.

Traditional healthcare management often depends on manual documentation. Such approaches require additional time, create storage and organization difficulties, and make it harder to access previous medical information when needed. A centralized digital healthcare management system provides an alternative approach by organizing healthcare information within a structured database and providing a common interface for managing records.

1.1 Project Background

Healthcare organizations handle large volumes of patient and administrative information. When this information is maintained manually or across disconnected records, maintaining consistency and retrieving historical information can become difficult. A computerized healthcare management system can centralize these records and support more organized healthcare administration.

1.2 Problem Statement

The project addresses the difficulties associated with maintaining healthcare information through manual or fragmented processes. These difficulties include record maintenance, slow information searching, risk of missing records, appointment management problems, and errors in billing and insurance information.

1.3 Project Motivation

The motivation behind the project is to develop a simple and effective healthcare management platform that reduces manual work and provides centralized access to healthcare information. The system is intended to demonstrate how a web-based application can organize multiple healthcare

activities within one platform.

1.4 Project Objectives

- Digitize patient healthcare records.
- Improve appointment management.
- Maintain complete medical history.
- Manage prescriptions and investigations.
- Automate billing and insurance management.
- Provide an easy-to-use healthcare dashboard.

1.5 Project Scope

The project covers a centralized web application containing the healthcare management modules implemented in the project. These include patient records, doctors, appointments, medical records, prescriptions, investigations, billing, insurance, symptom logs, and preventive care. The current implementation is designed as a web application using the technologies described in the technology stack.

Chapter 2

Problem Domain and Existing System

2.1 Existing System Analysis

Traditional healthcare management depends heavily on paper records and manual processes. Based on the project analysis, the major limitations are:

- Difficult record maintenance.
- Slow information searching.
- Risk of missing records.
- Appointment management difficulties.
- Billing and insurance errors.

2.2 Proposed System

The proposed Smart Healthcare Management System provides a centralized web application where healthcare information can be stored, updated, and accessed efficiently. The application groups multiple healthcare activities into dedicated modules while using a common MySQL database for persistent storage.

2.3 Advantages of the Proposed System

- Centralized management of healthcare information.
- Structured storage of patient and administrative records.
- Improved access to medical history and related information.
- Dedicated modules for appointments, prescriptions, investigations, billing, insurance, symptoms, and preventive care.
- Reduced dependence on manually maintained records.

Chapter 3

Requirement Analysis

3.1 Functional Requirements

The functional requirements define the healthcare management activities that the system is expected to support.

Table 3.1: Functional Requirements

No.	Module	Description
1	Patient Management	Manage patient information.
2	Doctor Management	Manage doctor profiles.
3	Appointments	Schedule and approve appointments.
4	Medical Records	Store diagnosis and treatment information.
5	Prescription	Manage medication details and prescription history.
6	Investigation	Store laboratory test information and results.
7	Billing	Generate invoices and manage payment status.
8	Insurance	Manage insurance policies and coverage information.
9	Symptoms	Track patient symptoms and severity.
10	Preventive Care	Store healthcare recommendations.

3.2 Non Functional Requirements

- Security using PDO prepared statements.
- Responsive user interface.
- Reliable database storage.
- Good performance.
- Maintainable modular design.

3.3 Technology Stack

Table 3.2: Technology Stack

Technology Area	Technology Used
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript
Backend	PHP
Database	MySQL
Database Communication	PDO
Server / Development Environment	Apache / XAMPP

Chapter 4

System Design

4.1 System Architecture

The system follows a three-layer architecture. Separating the presentation, application, and database responsibilities provides a clear structure for implementing and maintaining the application.

4.1.1 Presentation Layer

The presentation layer is responsible for user interaction. HTML, CSS, Bootstrap, and JavaScript are used to create the web interface through which users interact with the healthcare modules.

4.1.2 Application Layer

The application layer is implemented using PHP. It handles business logic and CRUD operations and acts as the application-level connection between the user interface and the database.

4.1.3 Database Layer

The database layer uses MySQL to store healthcare information. PDO is used for database communication and prepared SQL statements are included as a security measure.

4.2 Entity Relationship Diagram

The Entity Relationship Diagram represents the major database entities and their relationships. The Patient entity is the central entity, with related information maintained through medical records, prescriptions, investigations, billing, insurance, symptom logs, and appointments.

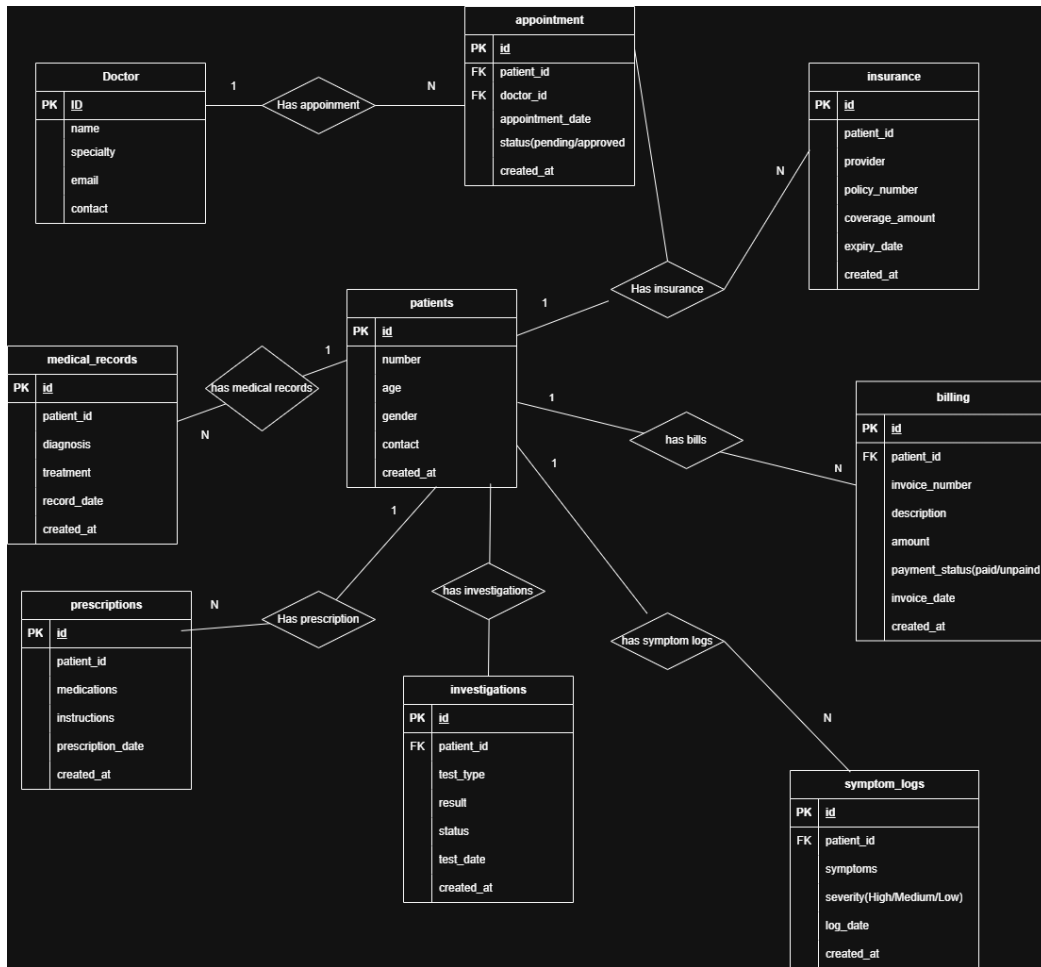


Figure 4.1: Entity Relationship Diagram of the Smart Healthcare Management System

4.3 Database Relationship Explanation

The Patient entity is the central entity of the database. The major relationships represented in the project are:

- One doctor can have many appointments.
- One patient can have many medical records.
- One patient can have many prescriptions.
- One patient can have many investigations.
- One patient can have many bills.
- One patient can have many insurance records.
- One patient can have many symptom logs.

Chapter 5

Software Development Life Cycle

5.1 Process Model

The Waterfall Model is used for this project. The development process is organized into sequential stages in which requirements are analyzed before system design, implementation, testing, deployment, and maintenance.

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

5.2 Application of the Process Model

The project follows the Waterfall sequence by first identifying the healthcare management requirements, then designing the architecture and database, implementing the modules, testing the system functionality, and considering deployment and maintenance activities.

Chapter 6

System Implementation

The SmartHealth dashboard provides centralized access to the healthcare modules. The implemented modules provide CRUD operations and dedicated interfaces for managing different categories of healthcare information.

6.1 Dashboard

Provides an overview of patients, doctors, appointments, and healthcare activities.

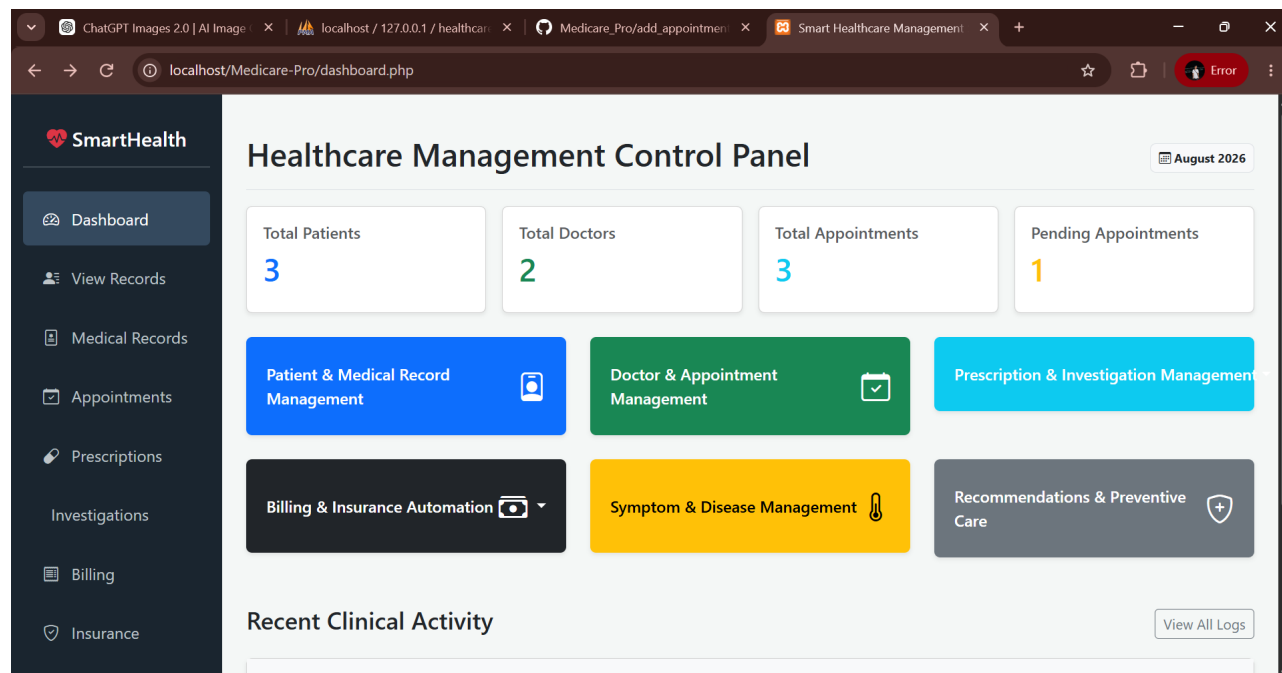


Figure 6.1: SmartHealth Dashboard

6.2 Patient Management

Allows administrators to add, view, edit, and delete patient information.

Patient Management [+ Add New Patient](#)

ID	Name	Age	Gender	Contact	Actions
3	S Ahmed	28	Male	1122334455	View Edit Delete
2	M Rahman	32	Female	0987654321	View Edit Delete
1	D Khan	45	Male	1234567890	View Edit Delete

Figure 6.2: Patient Management Module

6.3 Medical Records

Stores diagnosis and treatment history for patients.

Medical Records Management [+ Add Medical Record](#)

Medical record added successfully! ✕

Record ID	Patient Name	Diagnosis	Treatment	Date	Actions
4	M Rahman	blood pressure		2026-08-07	View Edit Delete
2	M Rahman	Migraine	Prescribed pain relievers	2026-07-28	View Edit Delete

Figure 6.3: Medical Records Management Module

6.4 Doctor and Appointment Management

Manages doctor information and the appointment approval process.

The screenshot shows the 'Doctor & Appointment Management' interface. On the left is a dark sidebar with the 'SmartHealth' logo and a menu including Dashboard, View Records, Medical Records, Appointments (highlighted), Prescriptions, Investigations, Billing, and Insurance. The main content area has a title 'Doctor & Appointment Management' and a '+ New Appointment' button. Below the title are two tabs: 'Appointments' (active) and 'Manage Doctors'. A table displays appointment data with columns for ID, Patient, Doctor, Date & Time, Status, and Actions.

ID	Patient	Doctor	Date & Time	Status	Actions
2	M Rahman	Dr. Jones	Jul 30, 2026 02:30 PM	Approved	No actions
1	D Khan	Dr. Smith	Jul 29, 2026 10:00 AM	Pending	Approve Cancel
3	M Rahman	Dr. Smith	Jul 03, 2026 06:02 PM	Approved	No actions

Figure 6.4: Doctor and Appointment Management Module

6.5 Prescription Management

Stores medication details and prescription history.

The screenshot shows the 'Prescription Management' interface. It features the same sidebar as Figure 6.4, with 'Prescriptions' highlighted in the menu. The main content area has a title 'Prescription Management' and a '+ New Prescription' button. A table displays prescription data with columns for ID, Patient, Medication, Dosage, Date, and Actions.

ID	Patient	Medication	Dosage	Date	Actions
1	D Khan	Amoxicillin	500mg	Jul 28, 2026	View Edit Delete
2	M Rahman	Ibuprofen	400mg	Jul 28, 2026	View Edit Delete

Figure 6.5: Prescription Management Module

6.6 Investigation Management

Maintains laboratory test information and results.

The screenshot shows the 'Investigation (Lab Tests) Management' interface. On the left is a dark sidebar with the 'SmartHealth' logo and a menu including Dashboard, View Records, Medical Records, Appointments, Prescriptions, Investigations (highlighted), Billing, and Insurance. The main content area has a title 'Investigation (Lab Tests) Management' and a '+ New Lab Test' button. Below is a table with columns: ID, Patient, Test Type, Result, Test Date, Status, and Actions. It contains two rows of data.

ID	Patient	Test Type	Result	Test Date	Status	Actions
1	D Khan	Blood Test	Normal	Jul 28, 2026	Completed	View Edit Delete
2	M Rahman	X-Ray	No abnormalities	Jul 28, 2026	Completed	View Edit Delete

Figure 6.6: Investigation Management Module

6.7 Billing Management

Creates invoices and manages payment status.

The screenshot shows the 'Billing & Invoices' interface. The sidebar is identical to the previous module. The main content area has a title 'Billing & Invoices' and a '+ Create Invoice' button. Below the title are three summary boxes: 'Total Invoices' (2), 'Total Paid' (\$200.00), and 'Total Unpaid' (\$150.00). Below these is a table with columns: Invoice #, Patient, Description, Amount, Date, Status, and Actions. It contains two rows of data.

Invoice #	Patient	Description	Amount	Date	Status	Actions
#000001	D Khan	Medical Services	\$150.00	Jul 28, 2026	Unpaid	Mark Paid View Edit Delete
#000002	M Rahman	Medical Services	\$200.00	Jul 28, 2026	Paid	View Edit Delete

Figure 6.7: Billing and Invoices Module

6.8 Insurance Management

Stores insurance policies and coverage information.

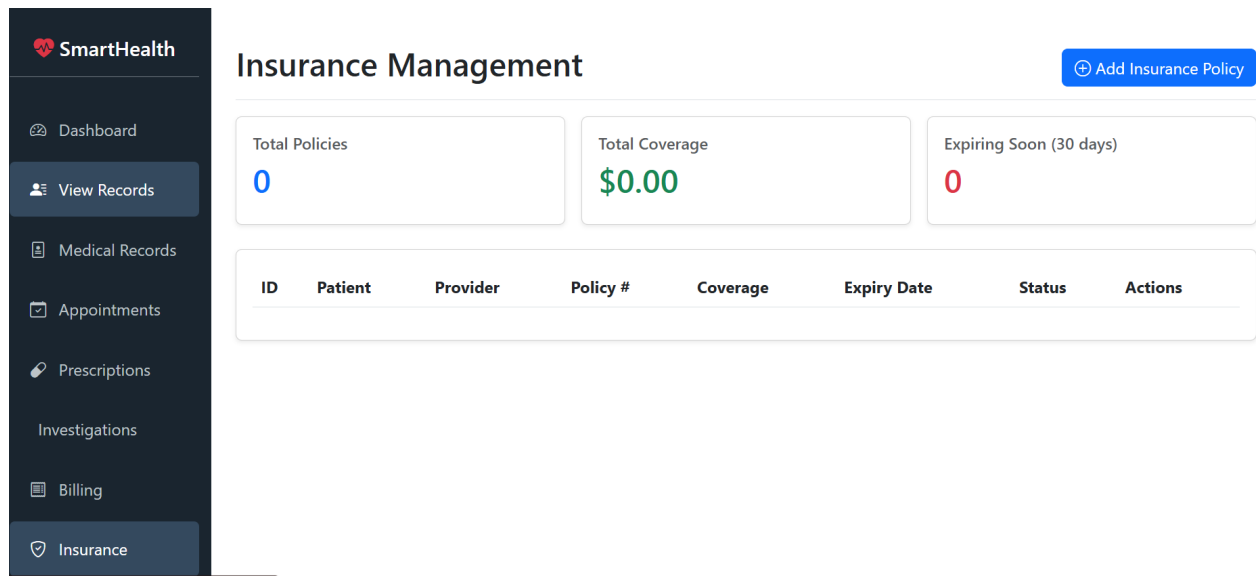


Figure 6.8: Insurance Management Module

6.9 Symptom Management

Tracks patient symptoms and severity.

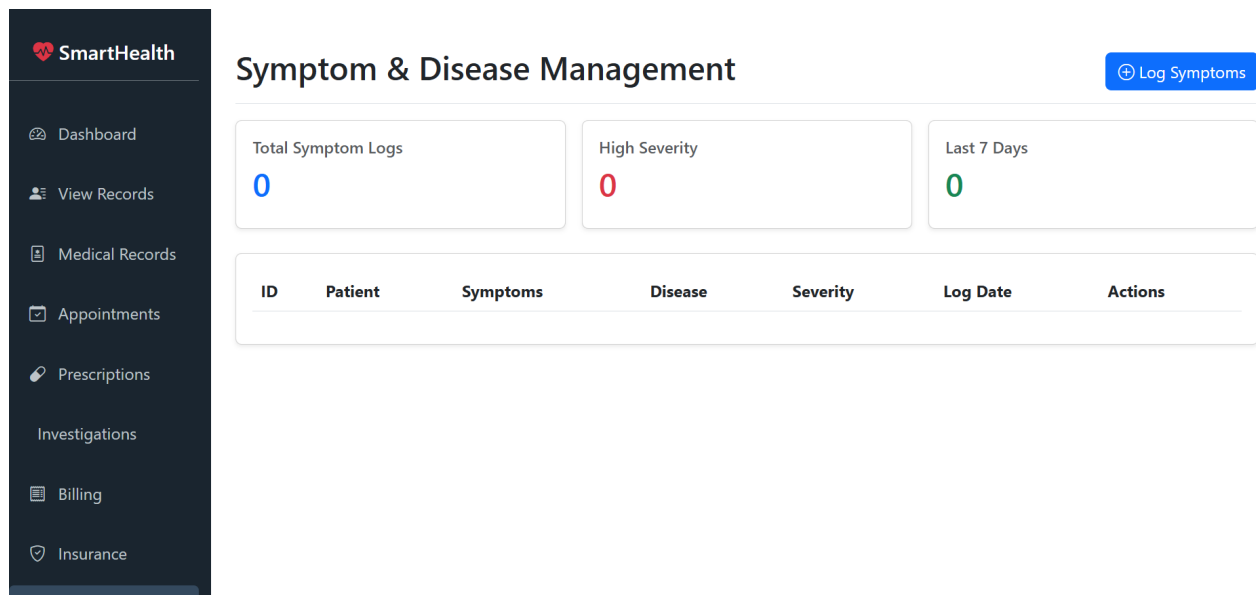


Figure 6.9: Symptom and Disease Management Module

6.10 Preventive Care

Stores healthcare recommendations.

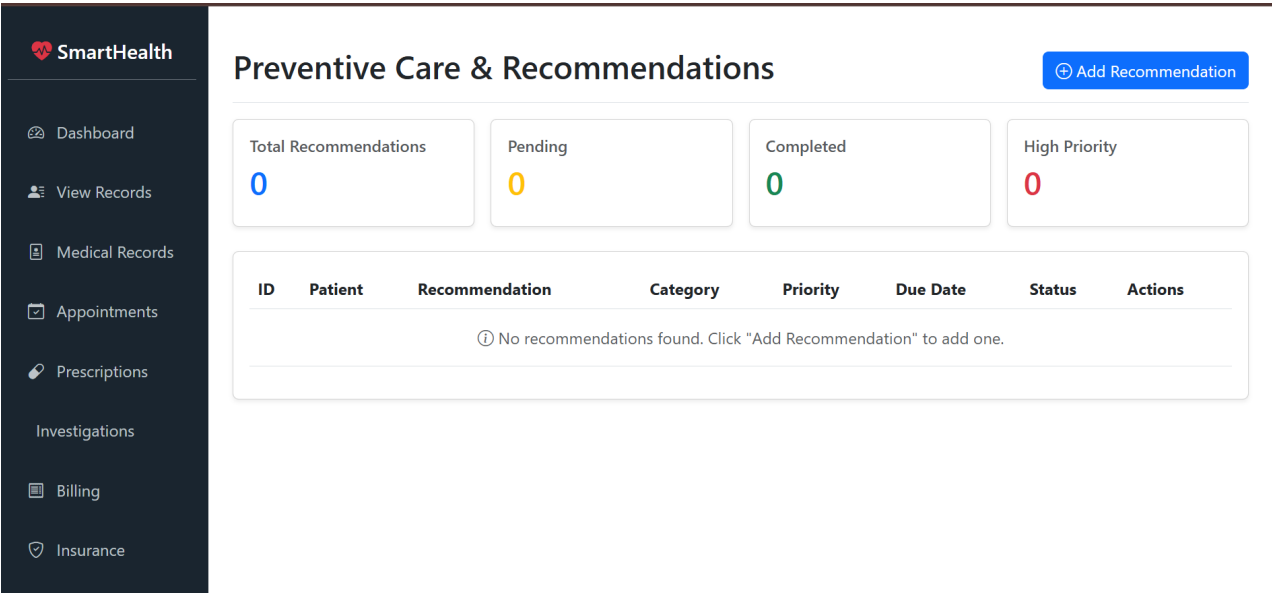


Figure 6.10: Preventive Care and Recommendations Module

Chapter 7

Testing and Security Analysis

7.1 Testing

Testing was performed to verify system functionality across the major healthcare management areas. The project identifies the following areas as part of the testing activity.

Table 7.1: System Test Cases

Test ID	Test Area	Purpose	Status
TC-01	Patient CRUD operations	Verify patient record create, read, update, and delete functionality.	Tested
TC-02	Appointment workflow	Verify appointment management workflow.	Tested
TC-03	Prescription management	Verify prescription information management.	Tested
TC-04	Laboratory investigation	Verify investigation information management.	Tested
TC-05	Billing operations	Verify billing and invoice operations.	Tested
TC-06	Database connectivity	Verify communication with the MySQL database.	Tested

7.2 Security Analysis

The system applies security measures including prepared SQL statements, input validation, and safe database processing. PDO prepared statements are used as part of the database communication approach. These techniques are intended to reduce database-related security risks and support reliable data handling.

Chapter 8

Results and Discussion

The implemented system demonstrates a centralized approach to healthcare information management. The dashboard brings the major healthcare functions into one interface, while individual modules provide dedicated screens for patient records, medical records, appointments, prescriptions, investigations, billing, insurance, symptoms, and preventive care.

The project also demonstrates the practical connection between the user interface, PHP application logic, and MySQL database. The inclusion of CRUD-based management across the modules provides a consistent approach to handling healthcare records.

The testing activities covered patient CRUD operations, the appointment workflow, prescription management, laboratory investigation, billing operations, and database connectivity. These activities were used to verify the functional areas identified in the project.

Chapter 9

Limitations

- No online payment gateway is currently implemented.
- No dedicated mobile application is currently provided.
- User role management is limited in the current implementation.

Chapter 10

Future Improvements

- Develop a dedicated mobile application.
- Deploy the system on a cloud environment.
- Introduce AI-based healthcare recommendations.
- Provide online consultation functionality.
- Add a notification system.

These improvements can extend the current web-based platform and provide additional ways for users to interact with healthcare services and information.

Chapter 11

Conclusion

The Smart Healthcare Management System provides an effective web-based solution for organizing and managing healthcare information. The system brings patient management, doctor and appointment management, medical records, prescriptions, investigations, billing, insurance, symptom management, and preventive care into a centralized platform.

The project demonstrates practical implementation of software engineering concepts including requirement analysis, system architecture, database modeling, implementation, testing, and security considerations. By organizing healthcare information within a structured application and database, the system reduces dependence on manual record management and provides a foundation for future enhancements such as mobile access, cloud deployment, AI-based recommendations, online consultation, and notifications.

Chapter 12

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